



IMAGE 1

(HCI Final Fall 2022 – Smartwatch Question)

◆ EXACT QUESTION (VERBATIM)

Q1:

In your class project you developed a transport application, which was primarily focused to work on a smart device or a web browser. Now you have been tasked to develop a version of the same application for a smartwatch. The size of the screen is 45mm and is connected through Bluetooth only. *(average time: 45 min, max marks 10)*

a) In your opinion, which TWO features of your original app (class project) you feel that you would like to add in the new smartwatch app. *(2 marks)*

b) Design a paper prototype (a sketch or a wireframe) to add these features. The prototype should have:

i. **User flows** – Follow identifying user flows. See how the users meet their goals and how they interact within the system.

ii. **Information entities** – Each user flow will show some user inputs and outputs. Identify what they are, how they relate to the user behaviour and expectations, what interactions they are involved with, and how they work?

iii. **Sketches** – Draw 2D sketches or wireframes. *(5 marks)*

c) Gestalt Principle of Proximity is that the relative distance between objects in a display affects our perception of whether and how the objects are organized into subgroups. How would you try to use this principle for your prototype. Explain by suggesting modifications to your prototype. *(3 marks)*

◆ WHAT THIS QUESTION IS ASKING

This is testing **three things**:

1. Can you **adapt functionality** to constraints (small screen, Bluetooth)?

2. Do you understand **prototyping properly** (flows, inputs/outputs, sketches)?
 3. Can you **apply Gestalt theory practically**, not define it.
-

◆ HOW TO THINK ABOUT THE ANSWER

- Smartwatch = **glanceable, fast, minimal**
 - Examiner wants:
 - **Justified feature choice**
 - **Clear user flow**
 - **Logical grouping of information**
 - **Explicit use of proximity**
-

✓ PERFECT EXAM ANSWER

(a) TWO features to add (2 marks)

I would add **real-time ride status notifications** and **quick ride confirmation/cancellation** to the smartwatch application. These features are suitable because they require minimal interaction, support glance-based usage, and allow the user to respond quickly without opening the mobile application.

(b) Paper Prototype Design (5 marks)

i. User Flows

The primary user flow begins when the user receives a ride request or status update on the smartwatch. The user taps the notification to view ride details such as ETA and driver information. The user then confirms or cancels the ride using a single tap, and the action is sent to the mobile application via Bluetooth.

ii. Information Entities

The main information entities include ride status, estimated arrival time, driver name, and confirmation options. User inputs consist of tap gestures for confirm or cancel actions. System outputs include visual notifications and confirmation

messages. These entities support quick decision-making and align with user expectations for minimal interaction on a smartwatch.

iii. Sketches

The prototype consists of three simple wireframes: a notification screen displaying ride status, a detail screen showing ETA and driver information, and a confirmation screen indicating successful action. Large buttons and minimal text are used to ensure readability on a small display.

(c) Gestalt Principle of Proximity (3 marks)

I would apply the Gestalt principle of proximity by placing related information such as ETA and driver details close together, while separating action buttons like confirm and cancel with sufficient spacing. This visual grouping helps users quickly understand which elements belong together and reduces cognitive load during interaction.



IMAGE 2

(Emotional Model + Social Media – E-commerce)

◆ EXACT QUESTION (VERBATIM)

Imagine, you have asked to influence an e-commerce site (eg. Amazon.com) to increase the sales by 100%.

To influence customers, you need to use some of the following features:

- i. Update the status on social media platforms.
- ii. Update the goal.
- iii. Set challenge parameters (date/goal/participation) and ask friends to join a challenge.
- iv. Set reward parameter.

a) What changes in your design you will suggest incorporating the combination of emotional model and social media features?

◆ WHAT THIS QUESTION IS ASKING

- Do you understand **Norman's Emotional Design Model**?
 - Can you link **emotion + social behaviour + design**?
 - Can you **apply theory**, not define it?
-

◆ HOW TO THINK ABOUT THE ANSWER

Mention:

- **Visceral** → visual appeal
 - **Behavioral** → ease of use
 - **Reflective** → pride, sharing, rewards
-

✓ PERFECT EXAM ANSWER

To increase sales, I would incorporate emotional design at all three levels. At the visceral level, I would enhance visual appeal using engaging product images and progress indicators. At the behavioral level, I would simplify goal setting and challenge participation to make interaction effortless. At the reflective level, I would enable users to share achievements and rewards on social media, creating social validation and motivation, which encourages repeat purchases and higher engagement.



IMAGE 3

(Amazon Book Section – Heuristic Evaluation)

◆ EXACT QUESTION (VERBATIM)

Q2 (b):

The development manager of a website for online book-buying section of Amazon.com has asked you to carry out a heuristic evaluation of the usability of the book section. He has specifically proposed the three heuristics listed below.

1. "There should be between five and nine navigation options on each page."
2. "There should be a good match between the navigation buttons and the users' goals."
3. "It should be easy for users to change their plans."

Based on your knowledge of the course, answer the following:

(a) Has the manager misunderstood heuristic evaluation? Briefly justify your answer.

(b) Please comment on the above three heuristics suggested by the manager. For each of the proposed heuristics, your comments should include:

- i. any theoretical justification for (or against) this heuristic;
- ii. any additional evaluation steps that might be required in applying it; and
- iii. the likely impact of such evaluation on the system design.

(average time 25 min, max marks 5)

◆ WHAT THIS QUESTION IS ASKING

- Do you understand **what heuristics really are**?
 - Can you **critique**, not blindly accept?
 - Can you link **theory** → **evaluation** → **redesign**?
-

✓ PERFECT EXAM ANSWER

(a) Misunderstanding?

Yes, the manager has misunderstood heuristic evaluation. Heuristics are general usability principles, not strict design rules or numerical limits. They guide evaluators in identifying usability problems rather than prescribing exact interface constraints.

(b) Comment on each heuristic

1. Five to nine navigation options

This heuristic is loosely based on cognitive load theory and Miller's Law; however, enforcing a fixed number of options is inappropriate. Evaluation would require observing user navigation behaviour. The impact may lead to redesigning navigation based on user needs rather than arbitrary limits.

2. Match between buttons and user goals

This aligns with Nielsen's heuristic of match between system and real-world expectations. Evaluation would involve task-based testing to verify alignment. The impact would improve clarity and task success rates.

3. Easy to change plans

This reflects user control and freedom. Evaluation would require testing undo and flexibility features. The impact would reduce user frustration and increase confidence in interaction.



IMAGE 4

(Clinical Genetics GUI – Family Tree)

◆ EXACT QUESTION (VERBATIM)

The following is a bespoke graphical user interface for software used in clinical genetics to assist with the collection of information about families.

In using the system, it has been noted that a number of errors have been made and you are tasked with planning a study to investigate these errors.

- a)** What are the main user related issues you might wish to investigate?
 - b)** What are the main human-computer interaction related issues you might want to investigate?
 - c)** Based on your analysis in a), b), and c), provide an outline of how you would evaluate the design using Nielsen Heuristic to improved design of the system.
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♦ WHAT THIS QUESTION IS ASKING

- Separate **human error** from **interface error**
 - Show you know **Nielsen's heuristics**
 - Explain **evaluation process**
-

PERFECT EXAM ANSWERS

(a) User-related issues

User-related issues include lack of training, misunderstanding of genetic symbols, cognitive overload, and data entry mistakes due to complex information representation.

(b) HCI-related issues

Human-computer interaction issues include poor visibility of system status, inconsistent iconography, lack of error prevention mechanisms, and cluttered interface layout.

(c) Nielsen Heuristic Evaluation

I would evaluate the system using Nielsen's heuristics such as consistency, error prevention, and recognition rather than recall. Identified violations would be documented and prioritized. Design improvements would then be implemented to reduce user errors and improve system usability.



IMAGE 5

(Quiz 5 – Short Questions)

♦ EXACT QUESTIONS & PERFECT ANSWERS

Q1: What are expressive interfaces? Write any two advantages of such interfaces.

Expressive interfaces are interfaces that communicate emotion or personality through visual, auditory, or interactive elements. Advantages include increased user engagement and improved user satisfaction.

Q2: Define Anthropomorphism. Where is this phenomenon commonly used? How it can be best used in HCI?

Anthropomorphism is the attribution of human characteristics to non-human systems. It is commonly used in chatbots and virtual assistants. In HCI, it is best used to improve trust, familiarity, and user engagement without misleading users.

Q3: Convert tasks into measurable usability tests

Example:

The user should successfully download the book within two minutes without external assistance.

(Repeat for each task using **time, success, accuracy**.)

Q4: What is meant by the carry principle in PCDs? What are its four important parameters?

The carry principle refers to the ease with which a personal communication device can be carried and used anytime. The four parameters are size, weight, battery life, and interaction speed.